

Major Project Report
On
Smart Footstep Power Generation Using Piezoelectric Sensors

Submitted in the partial fulfillment for the award of the
Degree of Bachelor of Technology

In
Electronics and Communication Engineering

by
MD. MUJAMMIL (22030133)

B.Tech. VIII Semester

Under the guidance of

Dr. BHANU P S DOHARE
(Associate Professor)



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SCHOOL OF STUDIES OF ENGINEERING AND TECHNOLOGY
GURU GHASIDAS VISHWAVIDYALAYA (A CENTRAL UNIVERSITY), BILASPUR (C.G.)

SESSION: 2025-26

ACKNOWLEDGEMENT

We are highly indebted to **Dr. BHANU PRATAP SINGH** for his guidance and constant supervision as well as for providing necessary information regarding the project and also for his/her support in the project. We owe our special thanks to our Head of Department **Dr. Sudhakar Singh Chauhan**, for encouraging us to acquire courage and knowledge through this project. We would like to express our gratitude towards our parents and faculty members of the Department of Electronics and Communication Engineering, School of Studies of Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur CG for their kind co-operation and encouragement which helped us in the completion of this project. We would like to express our special gratitude and thanks to all the staffs of Electronics and Communication Engineering department for giving us such attention and time. Our thanks and appreciations also go to our colleague in developing the project and people who have willingly helped us out with their abilities.

MD MUJAMMIL (22030133)

DECLARATION

We the under signed solemnly declare that this report of the major project work, entitled **Smart Footstep Power generation using piezoelectric sensors** is carried out during the course of our study during VIII semester under the guidance of **Dr. BHANU PRATAP SINGH**, Associate Professor, Department of Electronics and Communication Engineering, School of Studies in Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.). We further declare that this minor project work is presented for the partial fulfillment of the requirement of degree of Bachelor of Technology in Electronics & Communication Engineering, School of Studies of Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.). The results embodied in this thesis has not been submitted to any other university or institute for the award of any degree or diploma.

Date:

MD MUJAMMIL (22030133)

APPROVAL SHEET

This major project report entitled **Smart Footstep Power Generation Using Piezoelectric Sensors** by ,MD MUJAMMIL and BHAVIRIPUDI ABHISHEK is approved for the partial fulfillment of the requirement of the degree of Bachelor of Technology in Electronics and Communication Engineering.

Dr.BHANUPRATAPSINGH
(Associate Professor)

Examiners:
Signature

Date:.....
place: Bilaspur

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

**SCHOOL OF STUDIES OF ENGINEERING AND TECHNOLOGY GURU
GHASIDAS VISHWAVIDYALAYA, BILASPUR(C.G.)**

(A Central University established by the Central University Act 2009 No. 25 of 2009)



CERTIFICATE

It is certified that the major project entitled **Smart Footstep Power Generation Using Piezoelectric Sensors**, submitted by MD MUJAMMIL and BHAVIRIPUDI ABHISHEK in partial fulfilment of the requirements requirements of the award of the degree of Bachelor of Technology in Electronics and Communication

Engineering, School of Studies of Engineering and Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur, is carried out by them in the Department of Electronics and Communication Engineering during the session 2025-26

under the supervision and guidance of **Dr Bhanu Pratap Singh**, Associate Professor, Department of Electronics & Communication Engineering, School of Studies in Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur C. G.

Dr. Sudhakar Singh Chauhan

(Head of Department)

Department of Electronics & Communication Engineering

School of Studies of Engineering & Technology

Guru Ghasidas Vishwavidyalaya, Bilaspur CG

Abstract

With the increasing global demand for electricity and the rapid depletion of conventional fossil fuels, there is an urgent need for innovative, sustainable, and eco-friendly energy solutions. This project focuses on the development of a Smart Footstep Power Generation System that utilizes the piezoelectric effect to harness wasted mechanical energy from human locomotion and convert it into usable electrical energy.

The system is designed for high-footfall areas such as railway stations, shopping malls, and smart city infrastructure. It employs piezoelectric sensors embedded beneath walking surfaces that generate an electric charge when subjected to the mechanical stress and pressure of footsteps. To enhance the system's intelligence and utility, an Arduino-based microcontroller is integrated to monitor, regulate, and display the generated voltage and step count in real time. Experimental results from the prototype demonstrate that the system can successfully generate sufficient power to operate low-voltage applications, such as LED lighting and small electronic devices, while remaining cost-effective and maintenance-friendly. The project not only provides a practical solution to energy harvesting but also aligns with global sustainability goals by promoting the use of clean, decentralized, and renewable energy sources. Future scalability could see these systems networked across entire urban landscapes to contribute significantly to green energy generation.

Keywords: Piezoelectric Sensors, Energy Harvesting, Arduino, Renewable Energy, Smart City, Footstep Power Generation

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CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE OF THE PROJECT

The main objective of this project is to develop an innovative energy-harvesting system that captures and converts the mechanical energy produced by human footsteps into usable electrical energy using piezoelectric sensors. This project focuses on utilizing non-conventional energy sources in a smart and sustainable manner to meet the growing need for electricity, especially for low-power applications in public spaces and smart environments.

This system is designed to be both intelligent and efficient. By incorporating a microcontroller (Arduino), the system is not only able to generate energy from footsteps, but it can also monitor, regulate, display, and manage the generated electrical output in real time. This integration of hardware and smart electronics ensures maximum utility, real-time awareness, and control of the harvested energy.

Another important objective of the project is to create a prototype that is low-cost, scalable, and applicable in real-world scenarios such as schools, railway stations, malls, and smart city infrastructure. The project aims to demonstrate that even minimal, everyday human motion can become a valuable source of renewable energy if effectively harnessed and managed. It emphasizes sustainability, environmental conservation, and energy awareness while encouraging innovation in green technology.

1. To develop a smart energy harvesting system using piezoelectric sensors that converts footstep-induced mechanical energy into electrical energy.
2. To integrate an Arduino-based microcontroller for real-time monitoring, control, and display of the energy generated.
3. To design a prototype that is cost-effective, environmentally sustainable, and suitable for real-world implementation in high-footfall areas.
4. To store or utilize the generated energy to power low-voltage applications such as LED lighting or small electronic devices.
5. To promote the concept of non-conventional renewable energy and encourage innovation in smart energy systems for use in smart cities and educational environments.

6. To analyse the efficiency and practicality of piezoelectric materials in real-time energy conversion through experimental implementation.
7. To raise awareness about sustainable energy technologies by demonstrating a live working model that combines mechanical, electrical, and embedded systems.

1.2 GENERAL

The usage of traditional power generation methods, such as burning coal, wood, and diesel, is continuously depleting our natural resources, particularly fossil fuels. The demand for power has exceeded supply due to the rising population. In addition to this, traditional methods contribute to pollution.

The concept of Crowd Energy Farms has the potential to redefine urban spaces by adding a sense of fluidity and encouraging people to activate spaces with their movement. The Crowd Farm floor is composed of standard parts that can be easily replicated, but at this stage, production remains expensive. However, this technology could facilitate the future development of new urban landscapes, including athletic fields with spectator areas, music halls, theaters, nightclubs, and large gathering spaces for rallies, demonstrations, and celebrations. The proposal to utilize waste energy from foot power through human locomotion is

particularly relevant for highly populated countries like India and China, where roads, railway stations, bus stands, temples, and other public places are densely crowded with millions of people moving around the clock. Currently, this human bio-energy is being wasted, but if harnessed effectively, it could become a revolutionary source of sustainable energy.

Walking across a Crowd Farm floor can be both functional and engaging, allowing people to generate energy while walking, improving their health, and even earning incentives. The electrical energy generated from these farms can be used for nearby applications such as lighting, charging stations, and other power needs.

There are several alternative methods to generate electricity, and among them, footstep energy generation is an effective and sustainable approach. Walking is one of the most common human activities, and with every step, a person loses energy to the road surface in the form of impact, vibration, and sound. Due to the transfer of body weight onto the ground with each step, this energy can be tapped and converted into usable electrical power.

If such a device is embedded into footpaths, it can effectively convert foot impact energy into electricity, providing an innovative and eco-friendly solution to energy generation.

1.3 MOTIVATION FOR CHOOSING THE PROJECT

In today's world, the demand for electricity is increasing at an unprecedented rate due to rapid industrialization, urbanization, and growing dependency on electrical and electronic devices. At the same time, conventional energy sources such as fossil fuels are depleting, and their continued use has led to significant environmental problems including air pollution, global warming, and climate change. This scenario calls for innovative solutions that promote sustainability and harness renewable forms of energy.

The idea behind this project — generating electricity from footsteps using piezoelectric sensors — stems from a simple observation: millions of people walk every day in public spaces, schools, railway stations, airports, shopping malls, and even inside homes. These everyday movements involve a vast amount of mechanical energy that goes completely unused. If this mechanical energy can be captured and converted into electrical energy, even at a small scale, it could contribute to powering low-voltage applications such as lighting, sensors, or charging small devices. Over time and with proper scalability, such systems could reduce dependency on conventional power sources.

This realization served as a strong motivation for choosing this project. We were inspired by the possibility of contributing to clean energy solutions and making use of a non-conventional energy source that is typically overlooked. The piezoelectric effect, which allows certain materials to generate electricity under mechanical stress, offers a perfect mechanism for this purpose. Moreover, it represents a fusion of physics, electronics, and sustainability — three areas of great relevance in today's engineering landscape. Another key motivator was the opportunity to develop a

practical, real-world

application that is not just theoretical but demonstrable. This project combines hardware implementation with smart control using Arduino, enabling us to not only generate electricity but also measure and display it in real time. This adds educational value and helps visualize the concept of energy harvesting clearly. Additionally, the

project aligns with green engineering principles and supports the

United Nations Sustainable Development Goals (SDGs), particularly Goal 7:

"Affordable and Clean Energy." Through this work, we aim to promote awareness of renewable energy technologies and inspire further research and innovation in this field.

1.4 IMPORTANCE OF SMART FOOTSTEP POWERGENERATION USING PIEZOELECTRIC SENSORS IN OUR DAILY LIFE

The importance of smart footstep power generation using piezoelectric sensors in daily life lies in its potential to transform ordinary human activity into a valuable source of renewable energy. In a world facing serious energy challenges due to the depletion of fossil fuels and the environmental impacts of conventional power generation, the need for altern active, clean, and sustainable energy sources has become more urgent than ever. Piezoelectric technology offers a practical solution by utilizing the mechanical stress generated by footsteps to produce electricity—an approach that is both eco-friendly and highly innovative.

Every day, countless people walk on floors, stairs, pavements, and public spaces without realizing the untapped energy that each step carries. This energy, though small on an individual scale, becomes significant when multiplied by thousands or millions of steps in crowded urban areas such as railway stations, airports, schools, stadiums, and shopping malls. Footstep power generation aims to harness this wasted mechanical energy and convert it into usable electrical energy using piezoelectric materials, which generate an electric charge when subjected to pressure or vibration. These sensors are embedded beneath walking surfaces and, when stepped on, produce small amounts of electricity that can be stored or used immediately. What makes this system truly valuable in daily life is its integration with smart electronics such as microcontrollers (e.g., Arduino). This allows the system not only to generate power but also to manage, regulate, and display it in real time. For instance, an Arduino can monitor the voltage produced, store the data, control how energy is distributed, and power small devices or lighting systems as needed. This intelligent control makes the system efficient, adaptable, and interactive, adding immense value in smart city applications where data-driven energy management is crucial.

In everyday scenarios, such a system can power low-voltage applications like LED lights, digital signage, or public charging ports. It reduces dependence on the central grid, especially in high-footfall areas, and contributes to lowering electricity bills and overall energy consumption. In remote or off-grid locations where access to electricity is limited, piezoelectric footstep systems can provide basic lighting or emergency power, making a significant difference in people's quality of life.

Moreover, the system promotes awareness and education about renewable energy, especially among students and the general public. It serves as a live demonstration of how mechanical energy can be converted into electricity, sparking interest in science, sustainability, and green engineering. This not only enhances learning but also encourages environmentally responsible behaviour.

1.5 PROBLEM STATEMENT

With the continuous increase in population, the demand for electricity is also rising, due to excessive consumption of fossil fuels and other non-renewable resources. At the same time, a significant amount of energy is wasted in various forms, including human locomotion. Conventional power generation methods are becoming insufficient, and alternative solutions are needed to meet the growing energy demand sustainably.

Public places like railway stations, bus stands, temples, shopping malls, and other crowded areas experience high foot traffic daily. The kinetic energy exerted by human footsteps is currently wasted. However, if harnessed effectively, this energy could be converted into a usable form to generate electricity.

The Footstep Power Generation System proposes a solution by utilizing piezoelectric sensors or a rack and pinion mechanism to convert mechanical energy from footsteps into electrical energy. This generated power can be used for various applications, such as street lighting, home appliances, agriculture, and sensor-based systems in remote areas.

The key challenge is to design an efficient and cost-effective system that can effectively capture and convert footstep energy into electricity while ensuring durability and feasibility for large-scale implementation in highly populated areas. This project aims to address the energy crisis by providing an alternative renewable energy source through human locomotion.

There is also a gap in public awareness and educational exposure to the practical use of piezoelectric materials and micro-energy harvesting systems. Students and young engineers are often introduced to renewable energy through solar and wind projects, but seldom through piezoelectric technology, despite its simplicity and versatility.

Demonstration projects and real-world applications are needed to showcase the feasibility and benefits of such systems, especially in the context of smart cities, sustainable development, and eco-friendly public infrastructure.

Therefore, the problem this project addresses is multi-faceted: the over-reliance on non-renewable energy sources, the lack of intelligent and practical systems to harness footstep-based mechanical energy, the inefficient use of energy in public spaces, and the limited public knowledge and accessibility to small-scale energy harvesting technologies. The project aims to provide a practical, smart, and sustainable solution by developing a footstep power generation system using piezoelectric sensors integrated with an Arduino microcontroller for energy monitoring, management, and application. By implementing this system, the project intends to demonstrate how everyday human activity can be transformed into a source of clean, renewable energy.

1.6 SCOPE OF THE PROJECT

This project aims to design and implement a footstep power generation system using piezoelectric sensors to harness energy from human footsteps. The key objectives of this project include:

1. Design and Implementation of Piezoelectric Sensors:

- o The project will focus on integrating piezoelectric sensors into flooring systems to capture mechanical energy generated by human footfalls.
The sensors will be selected based on their efficiency in converting pressure
- o from footsteps into electrical energy.

2. Energy Harvesting Mechanism:

- o The system will convert the mechanical energy produced by footsteps into usable electrical power.
- o A power management unit will be developed to store and distribute the harvested energy, enabling its use in small electronic devices or to contribute to energy-saving applications in smart buildings.

3. System Design:

- o The project will involve designing a modular system that can be easily integrated into different environments such as shopping malls, airports, or residential buildings.
- o The design will emphasize energy efficiency, system scalability, and the feasibility of long-term operation.

4. Applications and Use Cases:

- o The project will explore the potential applications of footstep-powered energy, including its use in powering sensors, lights, or other low-power devices in public spaces.
- o The system could also be used to power smart infrastructure in urban planning or be incorporated into wearable devices that monitor human activity.

5. Environmental Impact:

- o The project will assess the environmental benefits of using renewable energy sourced from human movement.
- o It will contribute to the broader goal of reducing reliance on non-renewable

energy sources and minimizing carbon footprints in urban spaces.

6. Limitations:

- o The system's capacity will be limited to low-power applications. While it will be capable of harvesting sufficient energy for small devices, it may not be viable for large-scale power generation.
- o The implementation of piezoelectric-based energy harvesting may be constrained by factors like footstep frequency, sensor placement, and flooring materials.
- o The project will not cover advanced energy storage solutions beyond simple capacitors or batteries for low-power storage.

7. Future Scope:

- o The project will serve as a foundational prototype for further research in improving piezoelectric sensor efficiency, scaling up energy harvesters, and exploring more diverse applications.
- o Future work may involve integration with IoT (Internet of Things) systems for smart energy management or expanding the technology to generate energy from other types of human motion.

1.7 PROJECT OVERVIEW

The Footstep Power Generation System is an innovative approach to harnessing waste energy from human locomotion and converting it into usable electrical energy. With the rising demand for electricity and the depletion of non-renewable resources, alternative power generation methods are crucial. This project focuses on utilizing piezoelectric sensors or a rack and pinion mechanism to capture kinetic energy from footsteps and convert it into electrical power.

The system is designed to be implemented in high-footfall areas such as railway stations, bus stands, shopping malls, temples, and walkways, where millions of people move daily. As individuals walk over a specially designed energy-harvesting floor, the mechanical force from their steps is converted into electrical energy, which can be stored in batteries or directly used for applications such as street lighting, charging stations, home appliances, agricultural needs, and sensor-based systems in remote locations.

The project aims to provide a cost-effective, sustainable, and eco-friendly solution to the ongoing energy crisis by utilizing an untapped source of energy—human motion.

CHAPTER 2

LITERATURE SURVEY

2.1 Priyanka Baniya, Shivanjali Barge, Rahul Patil, Prof. Poonam Yewale[1]

The piezoelectric effect is the ability of certain materials to generate an electric charge in response to applied mechanical stress. In this project, the concept of piezoelectric power generation is explored, focusing on harvesting energy from vibrations and pressure caused by external forces. These forces may include human footfalls, moving vehicles, or other mechanical sources. The underlying idea is to convert mechanical energy, which is often wasted, into useful electrical energy using piezoelectric materials.

The proposed system employs piezoelectric elements placed in areas of regular mechanical impact. When mechanical pressure is applied to these materials, they produce an electric potential which can be collected, stored, and used for powering low-power electronic devices or charging batteries.

Research Gaps

Despite the promising potential of piezoelectric power generation, several research gaps remain:

1. **Low Energy Output:** Current piezoelectric materials generate low power output, which limits their application to small-scale devices. Research is needed to improve the efficiency and output capacity of piezoelectric materials and systems.
2. **Durability and Material Fatigue:** Long-term performance under repeated mechanical stress is not well-documented. The degradation of piezoelectric materials over time affects reliability and efficiency.
3. **Energy Storage Integration:** Effective integration with energy storage systems such as supercapacitors or rechargeable batteries is still in its early stages. More efficient power management circuits are required.
4. **Optimization for Real-World Applications:**

Most studies focus on theoretical models or small-scale prototypes. There is a lack of large-scale field testing, especially in dynamic environments like roads or public spaces.

5. **Cost vs. Benefit**

Analysis: While the concept is considered cost-effective, comprehensive economic evaluations and lifecycle cost analyses are limited in existing literature.

2.2 K. NARESH, A. BALAJI, M. RAMBABU, G. NAGARAJU[2]

With the increasing demand for portable electronic devices and the simultaneous emphasis on sustainable energy sources, energy harvesting from the surrounding environment has gained significant attention. One promising approach is the use of piezoelectric materials to convert mechanical energy—particularly from vibrations and pressure—into usable electrical energy.

piezoelectric materials generate an electric charge when subjected to mechanical stress. Conversely, they deform mechanically when an electric field is applied. This bidirectional property makes them suitable for both sensing and energy harvesting applications. Among various energy sources available in the environment, mechanical vibrations, such as those produced by human movement or vehicle motion, are ubiquitous and often wasted.

Piezoelectric films and elements can convert ambient vibrations and footstep-induced pressures into electrical energy. This energy can be stored in rechargeable batteries or capacitors and later used to power small electronic systems.

This principle finds useful application in public infrastructures such as airports, railway stations, and pedestrian walkways, where large volumes of foot traffic offer substantial kinetic energy. By embedding piezoelectric devices in flooring, it becomes possible to harvest this kinetic energy and use it to operate lighting systems or small displays, contributing to a reduction in the facility's grid energy consumption.

Research Gaps

While the application of piezoelectric energy harvesting is promising, several key research gaps need to be addressed to enable wider adoption and effectiveness:

1. **Power Output Limitations:** The electrical energy generated per footstep or vibration event is relatively low. More research is required to develop materials and configurations that significantly increase output power.
2. **Mechanical to Electrical Conversion Efficiency:** There is a lack of optimized designs for maximizing energy conversion efficiency from random, low-frequency vibrations such as footsteps.
3. **Durability and Long-Term Performance:** Piezoelectric materials often degrade over time when subjected to continuous stress. Studies on material fatigue and lifecycle performance are limited.

2.3 Kiran Bobby, Aleena Paul K, Anumol.C. V, Josnie Ann Thomas[3]

Human dependence on energy has continuously increased over time, driven by the need for comfort, development, and survival. However, this rising demand has led to the rapid depletion and wastage of conventional energy resources. In response to these challenges, energy harvesting from non-conventional and renewable sources, especially waste mechanical energy, is gaining prominence.

One such innovative method is the utilization of footstep energy, particularly relevant in densely populated countries like India. Public places such as railway stations, temples, and markets experience high volumes of foot traffic throughout the day. Harnessing this mechanical energy through piezoelectric flooring systems offers a practical and sustainable solution.

By embedding piezoelectric sensors or transducers into the flooring, the mechanical pressure exerted by footfalls is converted into electrical energy. These floor sensors detect

pressure variations and generate an electrical charge, which can be captured, stored, and later used as a power source.

Research Gaps The concept is promising, yet several research and development gaps need to be addressed for effective and widespread implementation:

1. **Limited Energy Density:** Footstep-based energy harvesting currently yields small amounts of energy per step, which may not be sufficient for high-demand applications without large-scale installations.
2. **Structural Design Optimization:** There is a lack of optimized floor designs that maximize energy output while maintaining durability, safety, and comfort for users.
3. **Cost-Effective Fabrication:** Development of low-cost yet efficient piezoelectric tiles and integration systems remains a challenge for large-scale deployment, particularly in developing countries.
4. **Energy Storage and Power Management:** Efficient systems are needed to store irregular energy inputs and regulate output for practical use in lighting or electronic devices.
5. **Data on Long-Term Reliability :** Real-world performance data on piezoelectric floors over long durations is limited, especially regarding wear and maintenance in high-traffic areas.

2.4 Namrata. Helonde, Punam Suryawanshi, Ankita Bhagwatkar, Arun Wagh[4]

Since ancient times, human society has relied on various forms of energy to support its growth, development, and daily activities. With increasing industrialization and population growth, energy demand has risen sharply, leading to the depletion and wastage of conventional energy resources. As a result, there is an urgent need to explore and develop sustainable, alternative energy solutions that utilize untapped sources.

One such potential lies in harvesting the mechanical energy generated by human locomotion—an everyday, yet largely wasted source of energy. This approach is especially relevant in densely populated countries like India, where places such as railway stations, religious temples, and public transit terminals experience continuous and heavy foot traffic. By implementing piezoelectric technology in flooring systems, it becomes possible to capture this kinetic energy and convert it into electrical energy.

When a person walks on a piezoelectric floor, the mechanical pressure exerted on the surface is detected by embedded sensors. These sensors, connected to piezoelectric transducers, convert the applied pressure into an electrical charge.

Research Gaps

Despite the promise of piezoelectric footstep energy harvesting, the following research gaps have been identified for future improvement:

1. **Power Generation Efficiency:** The actual energy output per step is limited. More studies are needed on advanced piezoelectric materials and configurations to improve energy conversion rates.
2. **Scalability and Installation Complexity:** Large-scale deployment (e.g., 100 floors or more) poses challenges in terms of system design, maintenance, and synchronization of output. Scalable and modular designs are still under development.
3. **Energy Storage Solutions:** There is a lack of integrated, efficient storage systems to handle intermittent and low-voltage outputs from footstep harvesting systems.
4. **Durability Under High Traffic:** The long-term durability and fatigue resistance of piezoelectric materials under constant foot traffic require extensive testing and optimization.
5. **Cost-Effectiveness and ROI:** The cost of implementing piezoelectric flooring must be weighed against the energy savings. Economic feasibility studies are limited.

CHAPTER 3

SYSTEM DESIGN & COMPONENTS

3.1 SYSTEM ARCHITECTURE

The overall system architecture of the footstep power generation project is designed to efficiently capture mechanical energy from human footsteps and convert it into usable electrical energy using piezoelectric sensors. The architecture consists of multiple functional blocks working in sequence to ensure energy harvesting, regulation, storage, and output delivery.

The system consists of the following key components:

1. Footsteps (Human Motion as an Energy Source) In our daily lives, thousands of people walk across public spaces, yet the mechanical energy generated from their footsteps is wasted. This project aims to tap into that unused potential. Here, the physical force exerted by human footsteps acts as the primary input to initiate energy generation. Every footstep becomes a source of renewable energy when integrated with the right technology. **2. Piezoelectric Sensor (Mechanical to Electrical Energy Conversion)** Piezoelectric sensors form the heart of this system. These sensors are made of crystalline materials that generate an electrical charge when subjected to mechanical stress. When a person steps on the sensor, the applied pressure causes deformation, which is instantly converted into an alternating voltage. Though small in magnitude, this voltage is valuable because it represents clean and renewable energy directly harvested from human movement.

3. Bridge Rectifier (AC to DC Conversion) The electrical output from piezoelectric sensors is alternating in nature and thus not directly usable for charging or powering DC devices. The bridge rectifier addresses this issue by converting the AC into DC. Constructed using four diodes in a specific configuration, the rectifier ensures that regardless of the polarity of the input, the output remains unidirectional, enabling further processing. **4. Capacitor (Energy Storage and Voltage Smoothing)** The rectified DC voltage is not smooth—it contains ripples that can affect downstream electronic components. To mitigate this, a 1000 μ F, 25V capacitor is introduced. It

performs two critical roles:

Energy Storage: Temporarily stores charge for future use.

Voltage Filtering: Smoothens the fluctuations in voltage, providing a steady output for the next stage. This capacitor acts as a short-term reservoir, ensuring reliability and stability in the energy flow.

5. Booster Module XL6019E1 (Voltage Amplification)

Since the energy from footsteps is typically low in voltage, it needs to be stepped up to usable levels. The XL6019E1 Booster Module increases the voltage to a range suitable for charging or further conversion. This module is essential in maintaining constant power output despite the irregular input from footsteps, making the system more efficient and reliable.

6. Charging Module TP4056 (Battery Management System) To store the generated energy for later use, we utilize rechargeable lithium-ion batteries. The TP4056 charging module is a dedicated battery management system that ensures safe and efficient charging. It protects against overcharging, overvoltage, and short circuits, significantly enhancing the safety and lifespan of the battery. This module acts as the bridge between energy generation and energy storage.

7. Output Booster Module (Regulated Power Supply)

Once the energy is stored, it must be supplied in a format compatible with standard electronic devices. The output booster module steps up the battery voltage (typically 3.7V) to a constant 5V. This regulated output is essential for powering USB devices such as smartphones, LED lights, or other portable electronics. It ensures compatibility and efficient energy delivery.

8. USB Port – 5V Output (User Interface) The final stage in this setup is a USB port that delivers a standard 5V output. This allows end-users to connect and charge their gadgets easily, just like they would with a regular power bank. The convenience and familiarity of the USB interface make this project not only technically functional but also highly practical for real-world usage.

3.2 SYSTEM ARCHITECTURE DESCRIPTION

The system architecture can be broken down into the following functional blocks:

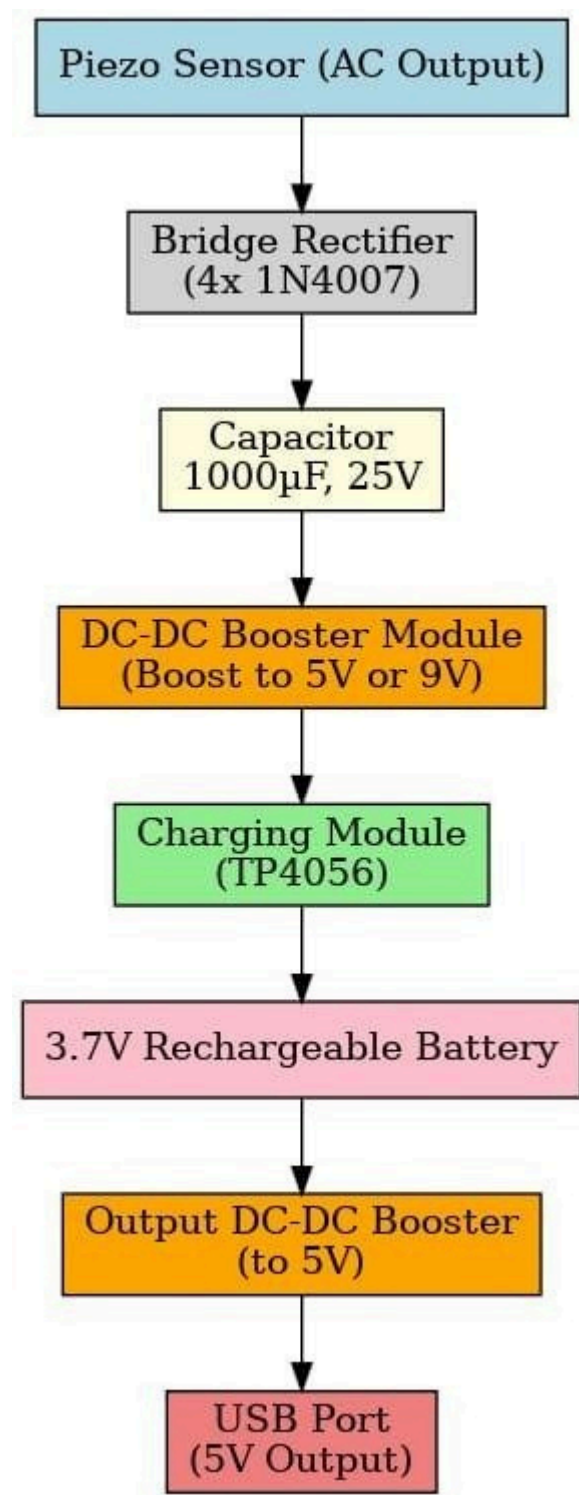


Fig No: - (i)

PIEZOELECTRIC SENSOR: -

3.3.DETAILED HARDWARE COMPONENT DESCRIPTION:

A piezoelectric sensor is a device that converts

mechanical stress (pressure, force, vibration, or strain) into an electrical signal using the piezoelectric effect. These sensors are widely used in applications like impact detection, vibration analysis, and medical devices. A piezoelectric sensor is a type of sensor that uses piezoelectric materials to convert mechanical stress, vibrations, or pressure into an electrical signal.

Here are some key features of Piezoelectric Sensor:

1. High Sensitivity: Piezoelectric sensors are highly sensitive to changes in pressure, acceleration, and Vibration.
2. Fast Response Time: They have a fast response time, making them suitable for detecting rapid changes.
3. High Frequency Range: They can detect frequencies up to several kilohertz.
4. Compact and Durable Design: These sensors are often small, robust, and capable of withstanding harsh environments including high temperature and pressure conditions.
5. Dynamic Response Only: They are best suited for dynamic (changing) measurements, as they cannot effectively measure static forces due to charge leakage over time.
6. Wide Frequency Range: Capable of detecting signals across a broad frequency spectrum.
7. Linearity: Maintains a linear relationship between mechanical input and electrical output.
8. Low Power Consumption: Ideal for energy-efficient systems and low-power electronics.



Fig No(ii)

DA IdOioDdEe :i s made up of two words i.e., “Di “means Two, and “Ode “means Electrodes which means that a device or component has two electrodes. (i.e., cathode and anode). A diode is an electronic device having a two-terminal unidirectional power supply i.e. it has two terminals and allows the current to flow only in one direction. Diodes are widely used in modern-day circuits to secure circuits from over-voltage and they are also used to change AC current to DC current.

Here are some key features of Diode:

1. **Unidirectional Current Flow:** A diode allows electric current to flow in only one direction — from the anode to the cathode and blocks it in the reverse direction.
2. **Threshold Voltage (Cut-in Voltage):** The minimum voltage required to turn the diode "on" in forward bias typically about 0.7V for silicon diodes and 0.3V for germanium diodes.
3. **Non-linear I-V Characteristics:** The current through a diode is not proportional to the voltage across it. Once the threshold is crossed, current increases exponentially.
4. **Breakdown Voltage:** The reverse voltage at which the diode starts conducting heavily, possibly leading to damage unless it's a Zener diode.
5. **Reverse Leakage Current:** A very small amount of current that flows in reverse bias, typically in microamperes.
6. **Low Forward Voltage Drop:** When conducting in the forward direction, a diode typically has a low voltage drop across it (e.g., ~0.7V for silicon diodes and ~0.3V for germanium diodes), allowing efficient current flow.
7. **Rectification:** Diodes are widely used in rectifier circuits to convert alternating current (AC) into direct current (DC).
8. **Switching Capability:** Diodes can act as electronic switches in digital and power electronics due to their ability to quickly switch between conducting and non-conducting states.



Fig No:-(iii)

ARDUINO UNO: - microcontroller that processes sensor data and controls the display. The Arduino Uno is a microcontroller-based board that functions by reading inputs (such as sensors, buttons, or data from other devices), processing that data using the programmed code, and then providing outputs (such as turning on an LED, running a motor, or sending signals to another device).

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Here are some key features of Arduino Uno:

1. Microcontroller: ATmega328P
2. Operating Voltage: 5V
3. Input Voltage: 7-12V
4. Digital I/O Pins: 14
5. Analog Input Pins: 6
6. Flash Memory: 32 KB (0.5 KB used by bootloader)
7. SRAM: 2 KB
8. EEPROM: 1 KB
9. Clock Speed: 16 MHz
10. Reset Button: Yes
11. DC Power Jack: Yes

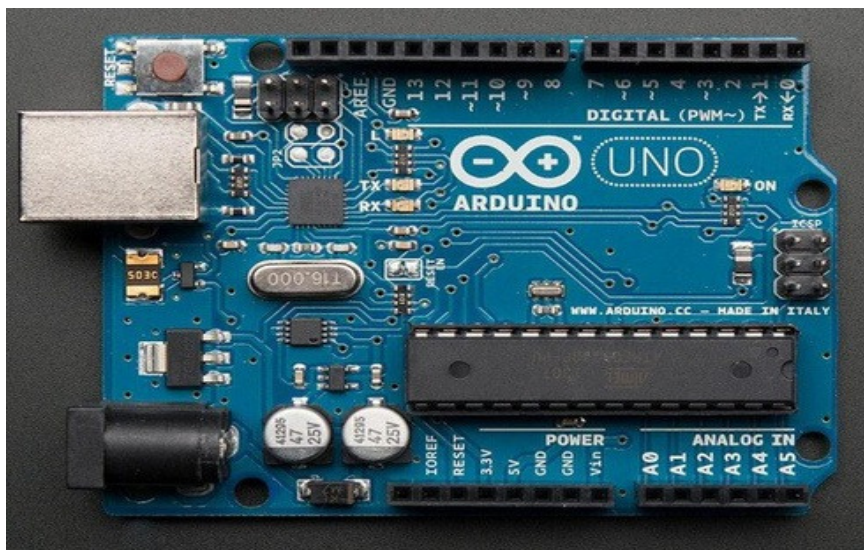


Fig No: - (iv)

LCD DISPLAY (16*2): - A 16x2 LCD display is a commonly used character-based display that can show 16 characters per row and has 2 rows. It is widely used in embedded systems, IoT projects, and industrial applications for data visualization.

A 16×2 LCD Display, more commonly known as a 16×2 alphanumeric Liquid Crystal Display (LCD), is a widely used electronic display module that consists of two rows with 16 characters per row, allowing a total of 32 characters to be displayed at once. Each character is displayed in a 5×8-pixel matrix, which enables the representation of letters, numbers, and a limited set of custom symbols. Though often referred to as LCD in casual terms, this display is technically an LCD (Liquid Crystal Display) that uses an LCD backlight for visibility.

The 16×2 LCD display is capable of displaying custom characters, blinking cursors, and scrolling text. It operates at a low voltage (typically 5V) and is energy-efficient. These displays are commonly used in embedded systems, DIY electronics, calculators, digital clocks, and many other appliances where text output is required.

Overall, a 16×2 LCD display offers a compact, cost-effective, and programmable way to show textual data, making it ideal for a wide range of applications in electronics and microcontroller-based systems.

Here are some key features of LCD DISPLAY:

2. . RDeissoplluatyio Sni:z 1e:6 Tcyhpaircaactleyr,s 3x- 42 ilnincehse s
3. Interface: Parallel interface (e.g., HD44780U)
4. Connection: Usually requires 4-6 wires to connect to a microcontroller
5. Character Size: 5x8 dot matrix per character
6. Contrast Adjustment: Via external potentiometer



FigureNo (v)

CAPACITOR: - A capacitor is a two-terminal passive electronic component that stores electrical energy in an electric field. It is composed of two conductive plates (typically metal) separated by a dielectric material (an insulating substance such as ceramic, plastic, glass, or air). When a voltage is applied across its terminals, electric charges of equal magnitude but opposite polarity accumulate on each plate, creating an electric field in the dielectric that stores energy.

The amount of electric charge a capacitor can store at a given voltage is measured in farads (F), the SI unit of capacitance. Most practical capacitors are measured in microfarads (μF), nano-farads (nF), or picofarads (pF), as one farad is very large for most circuits. The fundamental equation for a capacitor is $Q = C \times V$, where Q is the charge in coulombs, C is the capacitance in farads, and V is the voltage across the plates.

Capacitors are used in many electronic devices. They help smooth out power, block or pass certain signals, and are used in timing and tuning circuits. There are different types of capacitors like ceramic, electrolytic, and film, and each is used for different purposes based on size, cost, and function.

Here are some key features of Capacitor:

1. Energy Storage: Stores energy in the form of an electric field.
2. Filtering: Used to filter out unwanted frequencies in a circuit.
3. Coupling: Used to couple signals between stages of a circuit
4. Charge and Discharge Capability: Capacitors can charge (store energy) and discharge (release energy) very quickly, making them ideal for filtering and timing applications.
5. Voltage Dependent Behaviour: The voltage across a capacitor cannot change instantaneously, which is useful for smoothing voltage fluctuations in power supplies



Figure No (vi)

BOOSTER MODULE MT3608: - The MT3608 is a compact and efficient DC-DC boost converter module designed to step up lower input voltages to higher output voltages. It's widely utilized in electronics projects, especially those powered by batteries, where a higher voltage is required than what the power source provides.

This module can take an input voltage as low as 2V and boost it up to as much as 28V, depending on the input power and load conditions. It uses pulse-width modulation (PWM) switching to control energy transfer from input to output, which allows for high conversion efficiency and minimal heat generation in comparison to linear regulators.

The MT3608 module is small in size and can be easily mounted on a breadboard or soldered into a circuit. Its simplicity, versatility, and high performance make it a preferred choice in DIY electronics and low-voltage power supply design.

Here are some key features of BOOSTER MODULE MT3608:

1. Wide Input Voltage Range: Operates from 2V to 24V, making it suitable for use with batteries like Li-ion, AA, or solar cells.
2. Adjustable Output Voltage: Can boost the input to a higher adjustable voltage (up to 28V), set using a built-in potentiometer.
3. High Conversion Efficiency: Offers up to 93% efficiency, reducing energy loss and heat generation.
4. Low Quiescent Current: Consumes minimal current when in standby, ideal for battery-operated devices.
5. Compact Size: Small and lightweight, suitable for compact and portable electronics projects.
6. Ease of Use: Features clearly labelled input/output pins, onboard potentiometer for voltage adjustment, and simple integration into projects with minimal external components required.



Figure No (vii)

CHARGING MODULE TP4056 LITHIUM-ION RECHARGEABLE BATTERY:

The TP4056 is a small electronic module designed to safely charge single-cell 3.7V lithium-ion or lithium-polymer rechargeable batteries. It uses a constant current/constant voltage (CC/CV) charging method and is typically powered by a 5V source (like USB). A lithium-ion rechargeable battery is a type of battery that can be recharged and reused many times, commonly used in phones, laptops, flashlights, and portable electronics. It typically has a nominal voltage of 3.7V and a full charge voltage of 4.2V. Lithium-ion batteries are lightweight, have high energy density, and are widely used for their efficiency and long life.

The TP4056 is equipped with a micro-USB input, allowing easy charging via a standard 5V USB source, such as a power bank, computer, or wall adapter. The module includes a constant current (CC) and constant voltage (CV) charging algorithm, ensuring the battery is charged safely without overcharging or damaging the cells. The charging process consists of two stages:

Constant Current (CC): In the initial stage, the charging current remains constant as the battery voltage gradually increases.

Constant Voltage (CV): Once the battery reaches a specific voltage (typically 4.2V), the charger switches to constant voltage mode, reducing the current to safely finish the charging process.

Here are some key features of Charging Module TP4056 Lithium-Ion Rechargeable Battery: -

1. **Safe and Efficient Charging:** Uses a constant current and constant voltage charging algorithm, ensuring safe and efficient charging of 3.7V Li-ion batteries.
2. **Overcharge and Over-discharge Protection:** Protects the battery from overcharging (charging beyond 4.2V) and over-discharging, which can extend battery life and improve safety.
3. **Compact Size:** The TP4056 module is small and lightweight, making it ideal for integration into compact, portable electronics and DIY projects.

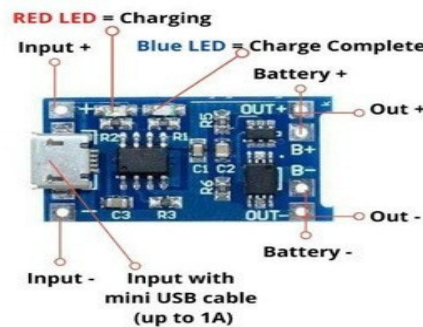


Figure No (viii)

USB FEMALE PORT: - A USB (Universal Serial Bus) Female Port is a physical interface socket designed to receive and connect with a USB Male connector. It serves as a standardized connection point for transferring data and electrical power between devices such as computers, mobile phones, flash drives, printers, cameras, and other peripherals. The female port is the receptacle side of the USB connection system, typically mounted on host devices such as computers, chargers, or power banks. It allows the user to plug in a USB male cable or device to establish a secure and functional connection.

These ports support plug-and-play functionality, which means that when a USB device is connected, the host system recognizes it automatically without needing a restart. USB female ports are essential in both data transmission (such as transferring files between devices) and power delivery (charging or powering electronic devices). In modern devices, USB female ports are also used for OTG (On-The-Go) capabilities, where one device can act as both host and peripheral.

Here are some key features of Usb Female Port:

1. **Standardized Interface:** Universally recognized and compatible with millions of devices worldwide, supporting both data and power connections.
2. **Receptacle for USB Male Connectors:** Serves as the socket into which USB male plugs are inserted to establish connections.
3. **Supports Data Transfer and Power Supply:** Used for both file/data transmission and charging or powering devices.
4. **Plug-and-Play Functionality:** Automatically recognizes and configures connected devices without manual driver installation in most modern systems.
5. **Hot-Swappable:** Devices can be connected or removed while the system is powered on, without risk of damage.

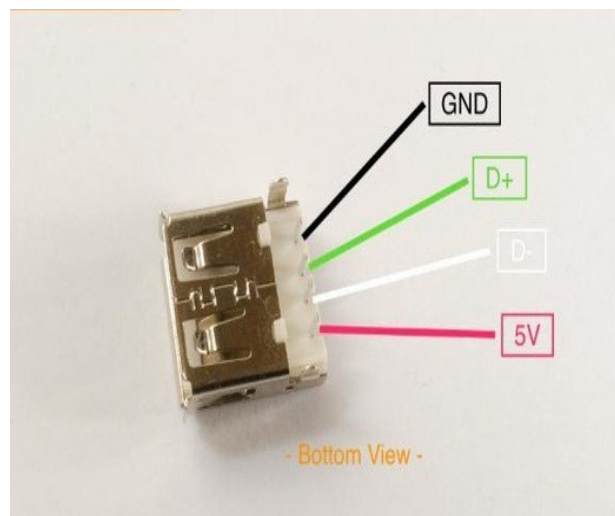


Figure No (ix)

RESISTOR: - A resistor is a passive electronic component designed to oppose or limit the flow of electric current in an electrical circuit. It is one of the most fundamental components in electronics, used in nearly every electrical and electronic device. The main purpose of a resistor is to control voltage and current within a circuit, ensuring that components operate within safe and efficient parameters.

Resistors operate based on Ohm's Law, which states that $V = I \times R$, where V is voltage, I is current, and R is resistance measured in ohms (Ω). By introducing a known resistance into a circuit, resistors can drop voltage, divide voltage, limit current to sensitive components like LEDs, and influence timing and filtering behaviours in combination with capacitors and inductors. Resistors come in many shapes, sizes, and types, including fixed resistors (with constant resistance) and variable resistors (like potentiometers or rheostats). They can be made from materials such as carbon film, metal film, wire-wound, or ceramic composition, each chosen for specific performance and application needs.

Here are some key features of Resistor: -

1. Resistance Value (Ohms, Ω): Defines how much the resistor resists the flow of current; available in a wide range from fractions of an ohm to several megaohms.
2. Power Rating (Wattage): Indicates how much power (heat) the resistor can safely dissipate without damage, typically ranging from 1/8 watt to several watts.
3. Tolerance: The allowable percentage variation from the stated resistance value, commonly $\pm 1\%$, $\pm 5\%$, or $\pm 10\%$, which affects accuracy in precision circuits.
4. Temperature Coefficient: Describes how the resistance value changes with temperature; important in applications where temperature stability is critical.
5. Type and Material: Comes in various types (carbon, metal film, wire-wound, etc.) and packages (through-hole or SMD), each suited for specific tasks based on performance and environment.
6. Non-polarized Component: Resistors can be placed in either direction in a circuit, simplifying installation



Figure No (x)

BREADBOARD: - A breadboard is a tool for prototyping electronic circuits, consisting of a grid of holes into which components and wires are inserted. Internally, the holes are connected by metal strips that create electrical connections. the breadboard gets its name from the original practice of building electronic circuits on literal wooden boards used for cutting bread, before specialized prototyping boards were invented. Today, modern breadboards are made from insulated plastic and include horizontal and vertical rows to help users create logical, organized circuit layouts.

Here are some key features of Breadboard: -

1. Solderless Prototyping: No soldering required, which makes it ideal for beginners and temporary circuit design.
2. Reusable and Flexible: Components can be inserted and removed multiple times, allowing easy experimentation and design changes.
3. Standardized Layout: Includes terminal strips and power rails with a typical 0.1-inch (2.54 mm) spacing, compatible with most electronic components.
4. Supports a Wide Range of Component: Can accommodate resistors, capacitors, transistors, ICs, LEDs, and jumper wires with ease.
5. Safe for Learning: Low risk of damaging components, making it perfect for students, hobbyists, and prototype testing.
6. Expandable Design: Multiple breadboards can be connected together for larger and more complex circuits.
7. Support for DIP Components: Designed to fit Dual Inline Package (DIP) ICs and components, commonly used in prototyping.
8. No Risk of Heat Damage: Unlike soldering, breadboards pose no risk of heat damage to sensitive components during assembly.
9. Versatile Power Rails: Equipped with side power rails for easy distribution of Vcc and GND across the board.

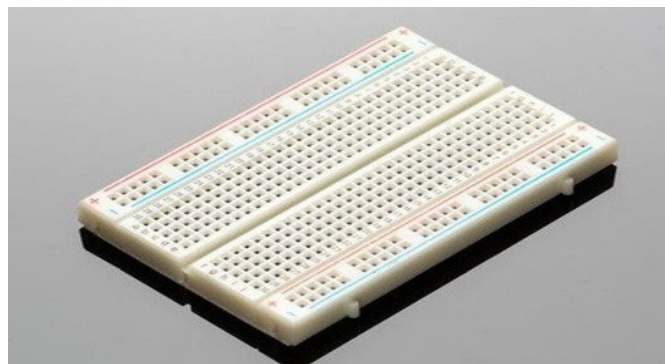


Figure No (xi)

JUMPER WIRE: - A jumper wire is a short, insulated electrical wire used to connect components in a circuit, especially on breadboards, development boards, and prototyping platforms. Jumper wires are essential for building and testing temporary circuits without soldering. They allow easy and flexible wiring between points on a breadboard or between a breadboard and external components like sensors, microcontrollers (e.g., Arduino, Raspberry Pi), or power supplies.

jumper wires are available in various lengths and colors, which helps in organizing and visually tracing connections in complex circuits. They are widely used in electronics education, robotics, embedded systems, and experimental circuit setups due to their plug-and-play nature and the ability to be reused multiple times.

Here are some key features of Jumper Wire: -

- 1.Reusable and Solderless: Ideal for temporary and experimental circuits; no soldering required, making them beginner-friendly.
- 2.Multiple End Types: Available in male-to-male, male-to-female, and female-to-female configurations to suit different connection needs.
- 3.Flexible and Durable Insulation: Covered with PVC or silicone that resists wear and makes wires safe to handle and bend without breaking.
- 4.Color-Coded for Easy Identification: Comes in various colours to help organize and trace connections in complex circuit setups.
- 5.Compatible with Breadboards and Headers: Standardized pin spacing (usually 2.54 mm) fits perfectly in breadboards, Arduino pins, and GPIO headers.
6. Available in Various Lengths: From short (10 cm) to long (30+ cm), allowing for different layout and spacing needs in circuit designs.
- 7.Compatible with Breadboards and Arduino: Designed to fit snugly into standard 0.1" (2.54 mm) pitch breadboards and microcontroller headers, such as those on Arduino boards.



Figure No (xii)

3.4 SOFTWARE DESIGN

software plays a supportive role in monitoring and controlling aspects of the hardware system, especially when integrating real-time data monitoring, display systems, or energy usage analytics. The software design is centred around microcontroller programming and simple interfacing with sensors and output devices. **1. Software Design Overview**

The software component of the footstep power generation system is responsible for:

- Reading analog voltage levels generated by piezoelectric sensors.
- Detecting and monitoring the system for faults.
- Logging data or transmitting it wirelessly (if IoT integration is used).

The design follows a sequential control logic programmed into the microcontroller, ensuring that each footstep event is accurately detected and processed in real time.

2. Programming Environment

The programming environment includes all tools and software platforms used for writing, compiling, uploading, and debugging the code.

a) Arduino IDE

- Purpose: Used for writing and uploading code to Arduino-compatible microcontrollers (e.g., Arduino Uno, Nano).
- Features:
 - User-friendly interface for code development.
 - Built-in libraries for analog reading, digital output, LCD, I2C, and serial communication.
 - Serial monitor for real-time data display and debugging.
- Programming Language: C/C++ (Arduino-based syntax)

b) Proteus (Optional for Simulation)

- Purpose: Simulates the circuit before physical implementation to test the interaction between hardware components and code.
- Features:
 - Real-time simulation of analog/digital circuits.
 - Supports virtual microcontroller loading.
 - Visualization of waveforms, voltages, and sensor output behaviour.

c) Fritzing (Optional for Circuit Design)

- Purpose: Used to create breadboard and schematic diagrams for documentation or PCB design.

- Features:
 - o Drag-and-drop component placement.
 - o Generates circuit schematics and layouts.

d) Arduino Libraries Used

- **LiquidCrystal.h:** For LCD interfacing.
- **Wire.h:** For I2C communication (if needed).
- **EEPROM.h:** For storing persistent values like footstep count.
- **SoftwareSerial.h:** For additional serial communication (e.g., with Bluetooth module).

3. Programming Logic

Below is a simplified overview of the logic implemented in the microcontroller:

1. Initialization:

- o Set pin modes for sensor input and display output.
- o Initialize serial communication and LCD.

2. Footstep Detection:

- o Continuously read voltage values from the analog input pin connected to the piezoelectric sensor.
- o Apply a voltage threshold to detect a valid footstep event.
- o Use debouncing or delay to avoid false triggers from vibrations.

3. Data Processing:

- o Increment footstep counts on valid detection.
- o Accumulate or calculate voltage output (optional).
- o Store values in memory or send to display.

4. Output Display:

- o Show footstep count and voltage on LCD.
- o Optionally transmit data via serial or Bluetooth.

3.5 CODE STRUCTURE

The embedded software code for this project was developed using the Arduino IDE in the C/C++ programming language. The code follows a modular and structured approach, ensuring readability, maintainability, and reliable real-time performance for step detection and voltage monitoring. Below is a breakdown of the main components and logical flow of the code:

1. Header Inclusions

All necessary libraries are included at the beginning of the code to enable access to specific hardware functionalities.

```
#include <LiquidCrystal.h> // For LCD display
```

2. Global Variable and Pin Definitions

Constants and global variables are declared to define pin connections and maintain state throughout the program.

```
LiquidCrystal lcd(7, 8, 9, 10, 11, 12); // LCD pin setup
```

```
const int sensorPin=A0;           //Analog pin connected to piezo sensor
const int threshold=100;          //Threshold for detecting a footstep
int footstepCount=0;              //Counter for number of steps
```

3. setup () Function

This function is executed once when the microcontroller starts. It initializes serial communication, LCD display, and sets up I/O configurations.

```
void setup() {
    lcd.begin(16, 2);           // Initialize LCD (16x2)
    Serial.begin(9600);         // Initialize serial communication
    lcd.print("Step Count: 0"); // Initial LCD display
}
```

4.loop()Function

This is the main loop that runs continuously, handling sensor reading, processing, and output display.

```
void loop() {
    int sensorValue = analogRead(sensorPin); // Read analog input

    if (sensorValue > threshold) {           //Check for valid footstep
```

```

    footstepCount++; // Increment step count

    //Voltage calculation
    float voltage = (sensorValue / 1023.0) * 5.0;

    //Display step count and voltage
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Steps: ");
    lcd.print(footstepCount);

    lcd.setCursor(0, 1);
    lcd.print("Volt: ");
    lcd.print(voltage, 2);
    lcd.print(" V");

    //Output to serial for logging/debugging
    Serial.print("Step ");
    Serial.print(footstepCount);
    Serial.print(" - Voltage: ");
    Serial.println(voltage);

    delay(500); // Debounce delay
}
}

```

Final Arduino Code for Smart Footsteps Power Generation System

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

//Set the LCD address to 0x27 or 0x3F (most common) for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2); // 0x27 is common, change to 0x3F if needed

int piezoPin = A0; // Piezo output connected to A0
int steps = 0;     // Step counter
float voltage = 0; // Voltage value

void setup() {
  lcd.init();      // Initialize the LCD
  lcd.backlight(); // Turn on backlight

  //Welcome message
  lcd.setCursor(0, 0);
  lcd.print("Footstep Power");
  lcd.setCursor(0, 1);
  lcd.print("Generation");

  delay(2000); // wait for 2 seconds

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Using Piezo");
  lcd.setCursor(0, 1);
  lcd.print("Sensor");

  delay(2000); // wait for 2 seconds

  lcd.clear(); // Ready for live display
}
```

```

voidloop() {
  intsensorValue = analogRead(piezoPin);

  //Convert analog reading to voltage
  voltage = sensorValue * (5.0 / 1023.0);

  //Ifvoltage exceeds threshold, count as a footstep
  if(voltage > 1.5) {
    steps++;
    delay(300); // debounce delay
  }

  //Display Steps and Voltage
  lcd.setCursor(0, 0);
  lcd.print("Steps:");
  lcd.print(steps);
  lcd.print("  "); // Clear extra digits

  lcd.setCursor(0, 1);
  lcd.print("Voltage:");
  lcd.print(voltage);
  lcd.print("V ");

  delay(500); // refresh every half second

}

```


CHAPTER 4

METHOD OF ANALYSIS OR PROBLEM FORMULATION

4.1 METHOD

The complete diagram of the power generation using footsteps. L-shapes window is inclined in certain small angle which is used to generate the power. The pushing power is converted into electrical energy by proper driving arrangement. The rack & pinion, spring arrangement is fixed at the footsteps which are mounded below the L shapes window. The spring is used to return the inclined L-shapes window in same position by releasing the load. The pinion shaft is connected to the supporter by end bearings. The larger sprocket also coupled with the pinion shaft, so that it is running the same speed of pinion. The larger sprocket is coupled to the small cycle sprocket with the help of chain cycle. This larger sprocket is used to transfer the rotation force to the smaller sprocket. The smaller sprocket is running same direction for the forward and reverse direction of rotational movement of the larger sprocket. This action looks like a cycle pedalling action

4.2 METHOD OF CONSTRUCTION

One of the major factors that determined the nature of the generation system was environmental issues. Constructing a device that generated power while conserving the environment was the most critical factor that motivated the idea of coming up with this generation system. The system is designed in a way that the people movement will be utilized to generate electricity. The footstep power generator basically translates the oscillatory motion to circular and later to electricity. The construction of the system includes measurement, manual cutting, drilling and welding.

4.3 MANUFACTURING ISSUE

Most of the material purchased did not conform to the measurement of the parts of the generator. Getting materials with similar measurements was impossible. In addition, some materials are not locally available. The last problem is the cost of the material. For instance, the price of steel is relatively high.

4.4 METHOD USED IN TO SOLVE THE PROBLEM

In order to get the correct measurement, measurement and manual cutting of the materials were done. Where the screws were needed, drilling was done to ensure that the bolts were fitted correctly. Other methods used in connecting different parts include welding. Welding was done where permanent attachment was needed. To ensure that enough time to make the cutting and measurement was available; all the materials were ordered in time.

4.5 CHALLENGE OF RURAL ELECTRIFICATION

The majority of the population living in developing countries does not have access to modern forms of energy and most of these two billion people are poor and live in rural areas. These rural dwellers rely on wood fuel for their cooking and heating needs and contrary to what is often thought most of them give priority to modern energy over many other necessities. Because of the lack of energy, including electricity, social and economic development is very limited or absent and living conditions remain at the low subsistence level. To foster rural development and adequate living conditions for the rural population both traditional and modern forms of energy will be needed.

It is generally accepted that there are well-performing utilities in the developing world, even in terms of electrification of rural areas. But there are many other developing countries where efforts to provide electricity to rural areas have only had limited success. Despite international support, many utilities continue to show substandard performance. As a result of recent technological advances, various alternative and lower cost options have emerged involving both traditional and renewable sources. These technological developments coincide with a growing concern for the ecological health of our planet and, in particular, the impact of emissions from the burning of fossil fuels. International and national agreements are forcing utilities to reduce emissions and, as a consequence, programmes are being implemented to increase the efficiency of the production and use of energy, and to promote the deployment of renewables.

Other important developments, which are mainly driven by neo-liberal ideology, are the recent regulatory changes and the introduction of competition into the electricity supply industry industrialised countries and in some developing countries.

These observations were conducive to starting the study on electricity supply to rural areas. The objective of the study was to identify and assess technical developments and institutional trends, to look ahead to opportunities for electricity supply to rural and

remote areas, and to translate the results into recommendations for decision makers. The study linked existing theoretical knowledge, practical experience and empirical findings, was approached from a utility perspective, and was marked by its width and multi-disciplinary character.

4.6 BENEFITS OF RURAL ELECTRIFICATION

Rural electrification is potentially a desirable investment in many countries but there has been considerable discussion about the socio-economic and environmental benefits, and the costs of the electrification of these areas in developing countries. The idea that rural electrification in itself substantially contributes to the development of rural and remote areas was already questioned in the eighties. On one hand this was prompted by the experience that most of the expectations were structurally not met, and on the other by a growing discomfort about the below standard performance of many rural power systems in developing countries. Demand forecast was often based on over-optimistic. It is noted that at the time most impact studies were of a descriptive nature but generally speaking there existed no acceptable correspondence between costs and benefits. There is general agreement that the cost and benefits of rural electrification are very difficult to value. But the results of ongoing research based on an advanced assessment of the benefit suggest that the socio-economic benefits of rural electrification are much higher than earlier assumed. For many rural people in the Third World however, electrification of their areas means modernity, progress and, above all, light in the darkness.

4.7 RESEARCH FINDINGS

The earlier research on electricity supply to rural areas has tended to address the technical and financial performance of both grid connected and decentralized power systems and the socio-economic impact of electrification. The implications of recent power sector development and trend for utility organization and management seem to have received less attention. In the present study rural electrification programmes implemented in both industrialized and developing countries, were analysed and conclusions drawn. The survey of a number of cases revealed that economic circumstances were not apparently a decisive factor in the wide-scale electrification of rural areas in industrialised countries. It is also fair to conclude that at a national level, in most cases, politicians have been far from proactive. The power of lobbies and pressure groups was probably a larger determinant. In many countries the rural population had to rise up against neglect before any actions on the electrification of their areas were taken. In general, the political leverage of the rural population in developing countries is low and this underlines the importance of a proactive and worldwide approach to the problems that rural populations

experience. There is a salient difference between presently developing countries and the industrialized nations around 1900. The latter already had reasonably developed economies and rural infrastructures, and they could afford to invest in electrification. In many European countries, electrification was considered to be more than merely an economic activity, the electricity supply to rural areas was seen as socially important. do not allow major investments in infrastructure.

As the historical analysis has shown, the electrification of rural areas should not be assessed as an independent activity, but rather as one of the components of an integrated rural development program. We should not ask ourselves whether electrification of certain areas is feasible, but what we are willing to spend for improvement of living conditions and rural development in general. There is also evidence that an integrated approach to rural development, including electrification, could lead to productivity and business activity synergies, provided that the energy service is reliable. Entrepreneurs in rural areas of the developing world consider unreliable and low service equality as the most important problem for their companies. In many countries, either a national or a provincial power utility was considered appropriate in order to avoid municipal boundaries hampering the development of an economically justified electricity infrastructure, and to achieve an acceptable area coverage and sufficient uniformity of tariffs. Small-scale private rural utilities have seldom proved to be successful (picture 1). In quite a few countries co-operatives have been successful, but success seems to be linked with the specific situation and culture in the country, and the opportunities for technical, administrative and managerial support. In many industrialised countries, and some developing countries, separate organisations were made responsible for the implementation of rural electrification programmes

These organisations were frequently decentralised offices of national utiliser, in some cases, specially established co-operatives. Semi-governmental monopolies proved to be a good solution in the power sector and the experiences in the industrialised world showed that this arrangement safeguarded the continuity of the utility and avoided neglect of financially less attractive supply areas. However rigid legislation and a bureaucratic attitude have often given monopolistic utilities the reputation of being non-participative and have hampered energy-efficient joint power generation with, for example, industry.

The experience in many countries shows that the performance of government-own Ed utilities can be satisfactory provided they are able to operate at an arm's length from the politicians and are reasonably autonomous. Such organisations have established and maintained a reliable and affordable electricity supply and have also demonstrated in other respects an appropriate performance. The ongoing reform of the power sector in industrialised countries is therefore not induced by a substandard performance of the utilities, but is rather the result of the prevailing neo-liberal ideology. Privatisation and

the introduction of competition in the power sector are seen as means to improve the efficiency relative to that in a monopolistic situation.

Under a liberalised regime the government is perceived as withdrawing from the power sector, but experience shows that strong regulation is needed to avoid the neglect of energy efficiency issues and the deployment of renewables, and to protect certain consumer groups. Government intervention is also required to control the monopolistic grids and the charges for the transmission of electricity.

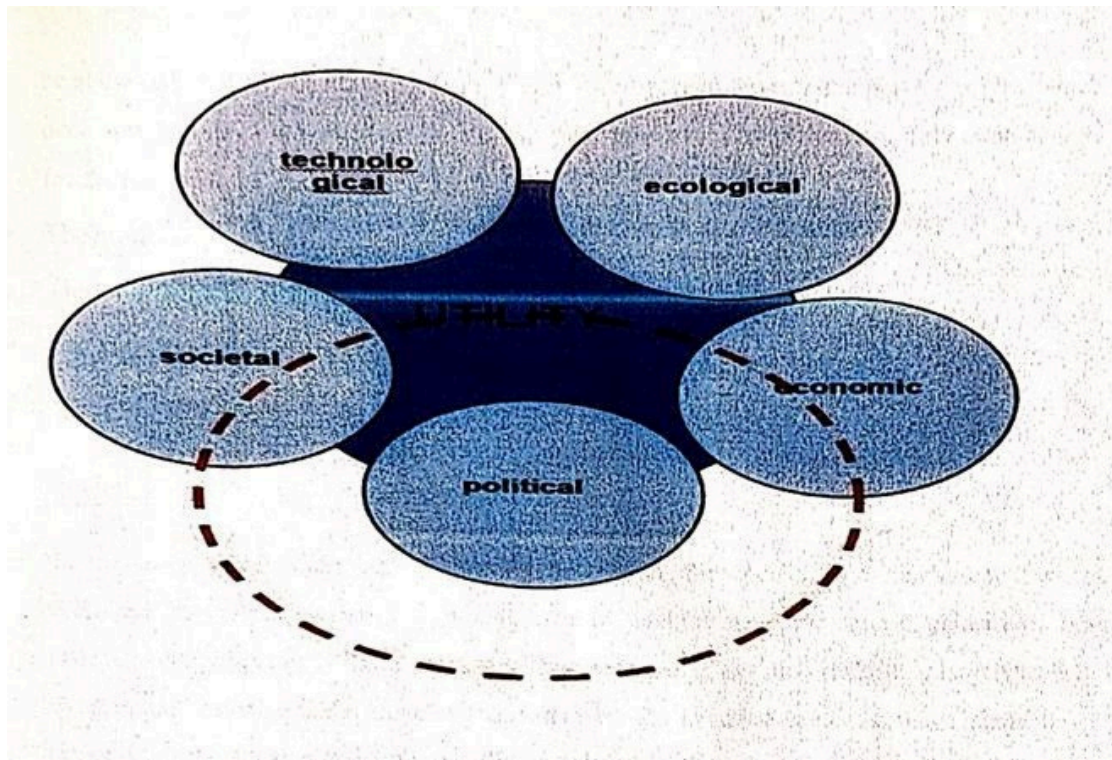


Figure No (xiii)

he unsatisfactory situation in several developing countries demands reforms to the electricity sector. However, the extent of any reform should be approached with "a healthy degree" of scepticism: the market is relatively small, the infrastructure is limited and expanding, and often the human resources needed for strong regulation are lacking. While industrialised countries could ask themselves the question what is the maximum amount of power sector reform that is desirable, the question for the developing regions could be what is the minimum reform needed to achieve proper working relations between the political elite and utility management. The electrification of rural areas has traditionally been based on electricity supply from a central grid. This preference emerged at the beginning of the last century due to the "economies of scale" achieved with large power stations. Decentralised schemes with, for example, diesel generators often led to relatively expensive electricity supply systems (picture 2). However, as a result of recent technological advances, various alternative options have emerged involving both traditional and renewable sources. These technologies that offer enhanced

opportunities for decentralized power.

As a result of recent developments, the electricity supply sector in many industrialised and developing countries faces unprecedented changes which affect both the sector itself and its customers. Current developments often refer to environmental, technological, societal and institutional issues and there is a growing consensus that the notion of "sustainability" should.

4.8 THE CHALLENGE AND SUCCESS FACTORS

The linkages between sustainable development and appropriate energy services have long been debated but it is now generally accepted that they are critical. Leaving fairness, equity and socio-economic development aside, the world can no longer continue to neglect, in terms of access to affordable electricity, the still unserved rural and remote areas. Progressive unification will gradually force the international community to satisfy the material needs of the global population, and energy, particularly electricity, will play an important role in this respect. The question will not be, are rural areas to be electrified, but rather when. As demonstrated by space research, mankind is technically and organisationally capable of releasing complex and impressive projects. We are capable of having a number of people remaining extra-terrestrial for many months. Further removed from us than the furthestmost rural dweller on this planet, the astronauts have an extremely reliable electricity supply at their disposal for food preparation, air- conditioning, heating, telecommunication, video equipment and scientific experiments. The power available on board the Space Station Alpha for housekeeping and medical care, could fulfil the basic electricity needs of a few hundred rural households in developing countries by using modern energy-efficient appliances. There is no reason to believe that extra-terrestrial power supply is economically more feasible, technically less complicated, or socially more obvious than the energy supply to the under developed rural and remote areas on earth. It is generally accepted that rural electricity supply is expensive relative to urban electricity supply and is has therefore often been considered by utilities as somewhat of a chore. But, continuation of the "business-as-usual" approach is not a solution for the two billion people still without access to electricity. In view of the currently available technological options and international community support rural electrification should be seen as a challenge. To succeed in this challenge and to successfully implement future rural electrification programmes the appropriate circumstances are needed and in this respect the study has revealed a number of

critical success factors:

A politically and socially stable environment.

Proper institutional conditions. Appropriate legislation and regulation to enable a commercial approach to power supply and utility operations at more than an arm's length from the political world.

Support from the international community. The majority of the rural population in developing countries are poor, but nevertheless willing to pay a substantial part of their income for reliable electricity. Despite this, subsidies on initial investments and appropriate financing schemes with smart subsidies for the beneficiaries will be needed.

An appropriate electrification process. To meet the demand for energy services in the most efficient, sustainable and environmentally and socially acceptable ways, the electrification process should consider all the local opportunities for power production. Electrification should preferably be a component of an integrated rural development program.

More so than in the past, the electrification of rural and remote areas in developing countries will be based on decentralised power facilities, including Solar Home Systems. The latter system is an attractive option for satisfying initial electricity needs. However, as well-being increases so does electricity demand with the consequence, that after time, a more powerful electricity supply will be needed. This evolution is one of the reasons why any rural electrification activity should be designed and implemented within a policy framework and in a well-planned manner.

Excess power generated by rural industrial enterprises can be used to support public electricity supply and small-scale independent power producers should be seen as fully fledged suppliers of electricity (picture 4). These observations require a close co-operation between the utility and rural customers, and thus a well-developed marketing and technical function.

A well-developed strategy. The strategy should be based on the importance of establishing close relationship with all stakeholders, energy efficiency measures, the deployment of renewables, and consideration of a "multifunction utility".

4.9 PROBLEM FACED BY POWER SECTOR

India's power sector is one of the key sectors which form the foundation of the growth of the country. Despite the fact that India has surplus energy, it is facing huge problems that serve as an obstacle for supplying electricity to all needy people.

- The troubles of power companies can be traced to structural issues such as the:
- Absence of meaningful price reforms
- Unreliable fuel supply and
- The unsustainable finances of public sector power distribution companies.

Stressed accounts

- The power sector, which has around 34 stressed accounts worth Rs 1.8 lakh crore, is the biggest worry for the banks since resolution through IBC will erode the true value of assets. This can lead to more bottlenecks in projects as was operations.
- The Standing Committee on Energy observed that development in the power sector has not been balanced.
- While de-licensing generation helped increased generation activities, the other segments (transmission and distribution) have not been given much attention.
- The Electricity Policy does not look into the issues around clearances, land acquisition, continuance of old and inefficient plants, instability in FSA policies, and other regulatory challenges and delays.

Cost of supply

- Electricity distribution companies (discoms) in some States (Seven States (Uttar Pradesh, Bihar, Odisha, Jharkhand, Assam, Rajasthan, and Madhya Pradesh)) are already highly indebted. Their debts account for 17% of the accumulated liabilities
- Despite continued State subvention (except by Odisha), all these discoms have been consistently.
- running at a loss, accounting for about 47% of the loss in electricity distribution business Existing subsidized lifeline tariffs in these States appear unaffordable to the poor and certainly higher than in States with universal (or high) access.
- The average plant load factor (PLF) across the country has fallen to a multi-decadal low of 58%. This implies inefficiency and extra burden of fixed costs.

4.10 PRESENT STATE & FUTURE OUTLOOK OF ELECTRICITY GENERATION IN INDIA

Electricity infrastructure and production are important for a developing economy like that of India, which with a population of 1.2 billion and an area of 3.29 million km², is the 7th largest country in the world. With a GDP of US\$ 2.3 trillion in 2015 and an average GDP growth of 7% per year, the growth of the electricity sector will be important to sustain the economic output of the country. Total electricity generation in India during 2015 was 1300 TWh from both utilities and non-utilities.

In 2013, the electricity consumption from all sectors was 824 TWh, and with an average growth rate of 9%, and it was estimated to be approximately 980 TWh in 2015. The electrical network suffers from transmission losses of approximately 25%. In 2015, per capita electricity consumption was 746 kWh Since per capita electricity consumption has a positive relation with GDP per capita (eg, [4-6]), it can be used as a standard for judging the stage of economic development. Most countries with a GDP per capita of more than US\$ 10,000 have an electricity consumption of more than 4500 kWh per capita There are few exceptions to this case, depending on the structure of the economy in that country. It can be expected that with the growth of the Indian economy, the GDP per capita will improve, and hence, there will be a need for more electricity in the future.

4.11 Electricity Generation in India

Electricity production in India is mostly achieved through coal thermal power plants. Although there have been efforts to diversify the options, particularly in the case of renewable energies, coal remains the dominant source of electricity in the country. Since 2000, the share of electricity production from coal has been slowly increasing; it was 68% at the start of the millennium and has increased to 73% in 2013. The percentual shares of all other energy sources, except renewables, have decreased during that time. In 2015, India generated a total of 1078 TWh from coal, natural gas, oil, nuclear and hydropower sources.

Renewable and alternative electricity in the form of solar, wind, biomass and small hydropower (less than 25 MW) plants are also making big progress. These types of electricity source were estimated to generate approximately 70 TWh of electricity in 2015. Non-utilities or independent power producers have also been growing a rate of 9% over the years, and it was estimated that they produced approximately 185 TWh in 2015. India also imported some electricity from Bhutan in 2015 (5 TWh) to satisfy its demand.

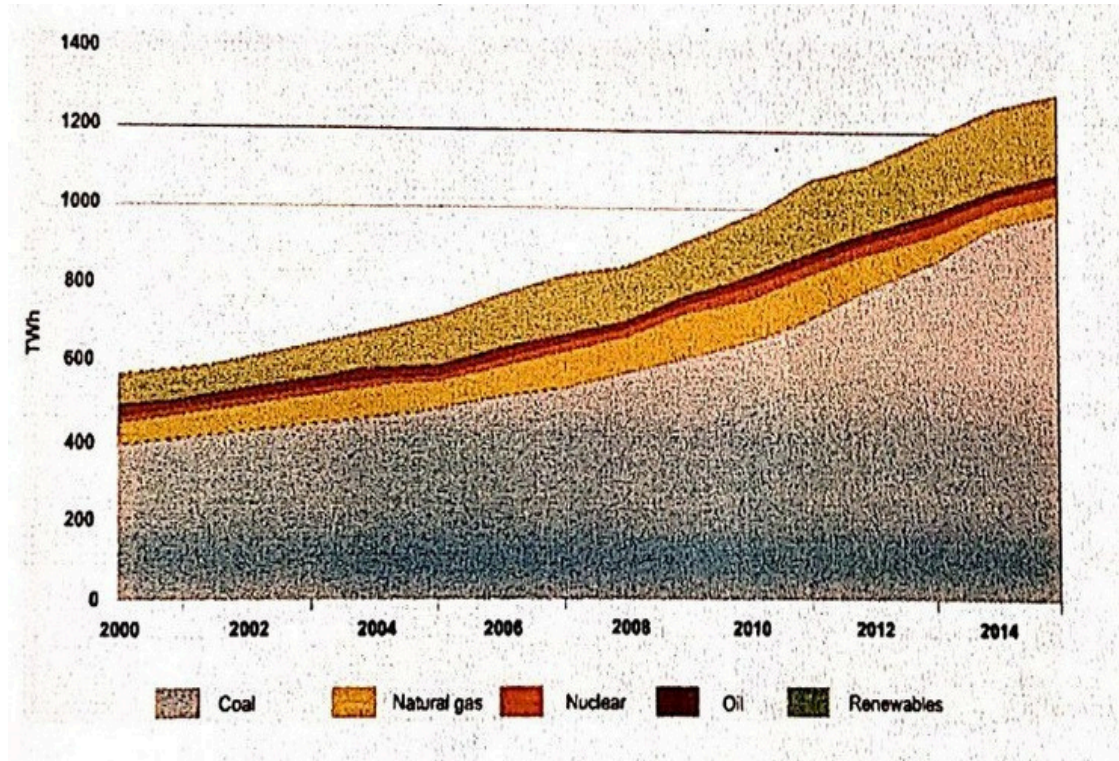
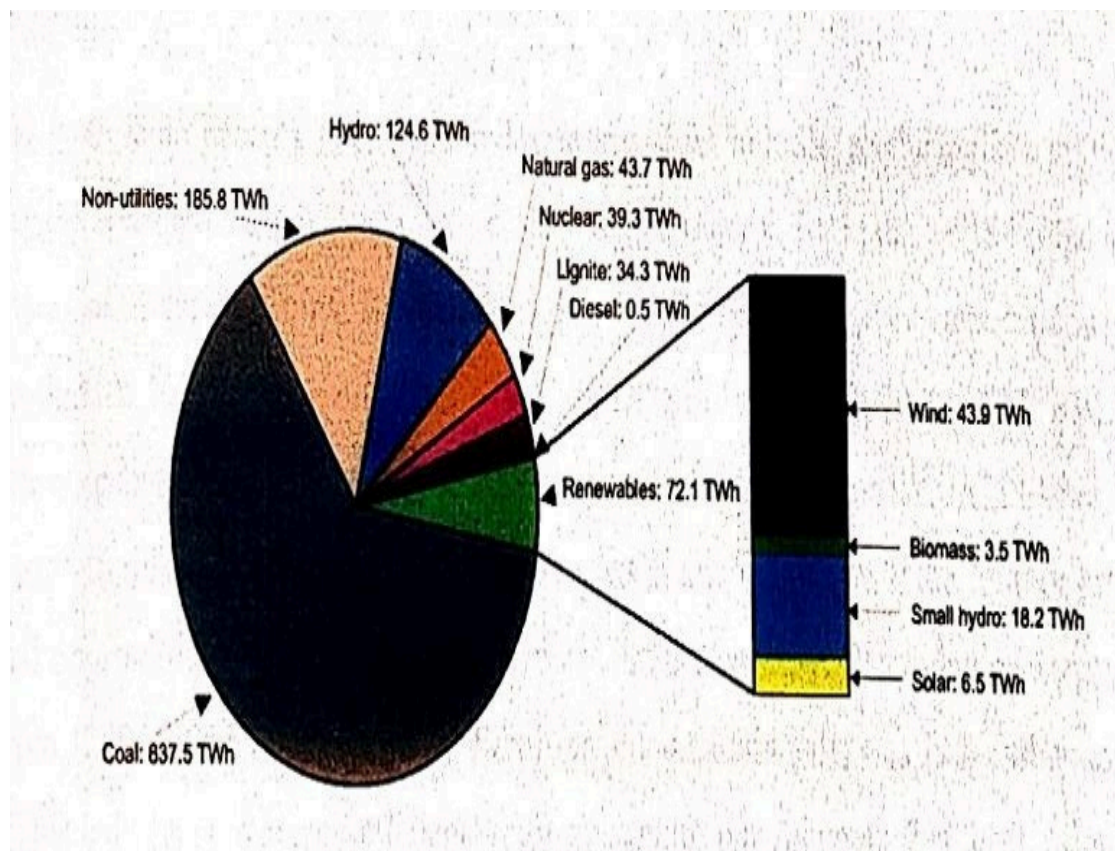


Figure No (xiv)

As mentioned, the bulk of the production comes from coal power plants that produced 840 TWh in 2015, which is 63% of the total production, followed by non-utilities (185 TWh) and hydropower (125 TWh, excluding small hydro). Nuclear, natural gas, lignite and oil also contributed 38 TWh, 44 TWh, 34 TWh and 500 GWh, respectively, to electricity generation. Renewable sources were estimated to contribute to 70 TWh of electricity in 2015. Currently, most of the renewable electricity generation is from wind energy and small hydropower plants. India is also keen on developing its renewable sources. In India, coal satisfies 70% of the total demand that is mostly used in the powersector, and with India having the 5th largest coal reserves in the world, it has proven to be the most economical form of energy and electricity in India. Considering an increase in coal production of 5% each year, the coal reserves are expected to last for another 40-50 years. The World Institute of Sustainable Energy prepared a research report for coal electricity in India and predicted that the total coal power plant capacity by the end of 2032 would be 400 TW if coal-based power plants are favoured or 220 TW if renewable energy and gas are favoured.



FigureNo (xv)

CHAPTER 5

RESULT AND DISCUSSION

5.1 EXPECTED RESULT

Power generated by foot step is designed and developed successfully, for the presentation purpose a prototype module is constructed with lower ratings of device, and results are found to be satisfactory and economical.

The initiative of electric power generation without polluting the environment. The waste energy in the form of human walking is utilized in this prototype. It is mainly useful at crowded place to install this system to produce electricity. This system is similar and less noisy in operation and provide greater flexibility in working. This system plays an important role for producing electricity at places where there are no sources of electricity like village areas.

If we install this model into the mat then expected power output maybe as below.

Let,

Mass of pedestrian = 65kg Area of plate = $10 \times 13 \text{ cm}^2 = 130 \text{ cm}^2$

Work done on plate by impact = weight of body distance $659.81 \times 0.1 \text{ Nm} = 63.765 \text{ J}$

So,

Power output = $\frac{\text{Work done}}{\text{time}} = \frac{63.765 \text{ J}}{1.06275 \text{ s}} = 60 \text{ Watts}$

It is like we have used only 3 piezo plates in parallel connected at the installed mat and we are getting likely 1 watt power output at only 10 cm size of plate.

As we know that, our cell phone needs 3-5V to charge the battery.

5.2 APPLICATION

- i. Foot step generated power can be used for agriculture, home application, street lightning.
- ii. Foot step power generation can be used in emergency power failure situations.
- iii. Metros, Rural application.
- iv. It can be used as a source for both AC and DC applications.
- v. It is also move in universities.
- vi. It can use in emergency power failure situations like hospitals.
- vii. Can be broadly utilized as a part of colleges, schools and universities.
- viii. This can be actualized in air terminals, transportations, rail road stations.
- ix. Street lights can be actualized utilizing this strategy.
- x. This framework can be actualized in swarmed places like shopping centers, pathways and so forth.

5.3 DESIGN EVOLUTION

The previous methods of power generation require a lot of capital to set a stable and reliable power generation system. In addition, some of the methods have various negative effects on environment such as air pollution, noise pollution among others. For this reason, coming up with a system that is ecosystem friendly was a nice idea. Therefore, the thoughts of coming up with the best power generation system that is silent, cost efficient, eco-friendly and reliable led to evolution of Footstep Power Generation System that meets all the above requirements.

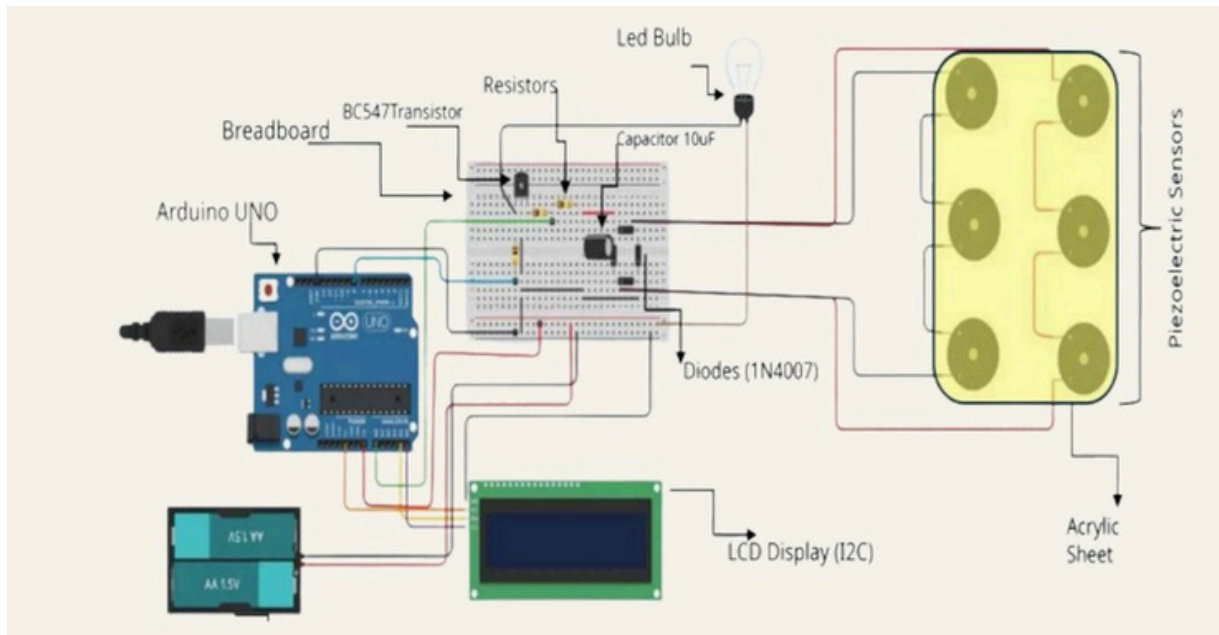


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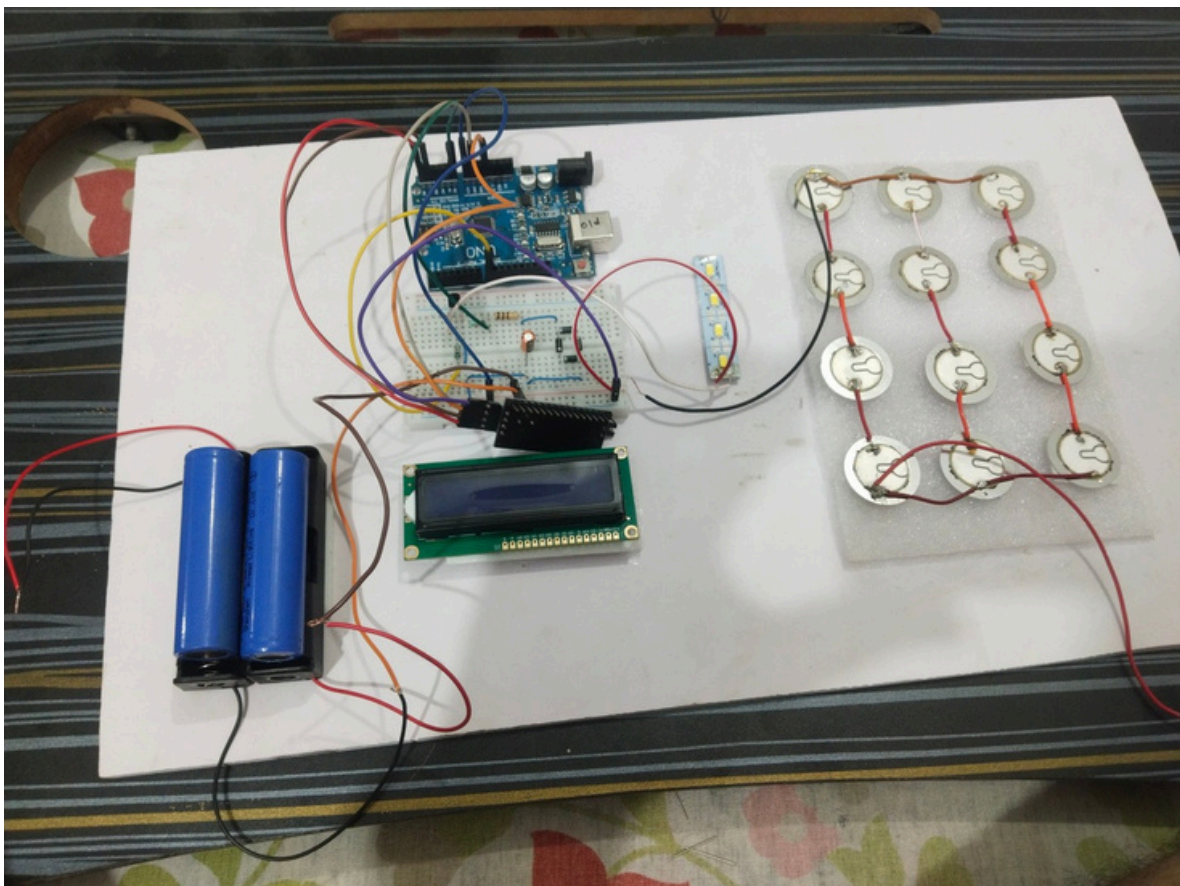
5.4 PROJECT RISK ANALYSIS

The system is risk free. Faulty of the system does not have any risk. The effect of the faulty on the system is only failure to generate power. The system is environmentally friendly and safe to use. The system is dependent on human movement. Therefore, the system depends on human movement which is renewable and implies that the system is only limited to work only on places with significant movement of people.

For security purposes during the construction of the system the following protective gadgets will be necessary gloves, dust mask, eye protection, the welding mask, footwear, hearing protection and protective clothes.

A power harvesting system is designed and implemented using piezoelectric sensor and voltage is harvested from pressure applied. Al the voltage status is displayed using LCD. Depending on the amount of pressure applied on the piezoelectric

sensor, a good amount of voltage is generated which can be used to charge e-vehicle. A switch is provided in the model by pushing the switch ON a bulb glow from the electric power obtained. We can display both the amount of electricity generated by the sensors and the energy stored in the battery in terms of voltage on 16x2 LCD. The main components of the system include piezoelectric transducers, ac ripple neutralizer, 8051 microcontroller, rechargeable battery, voltage sampler, ADC, LCD display and kept for charging E-vehicles. Here in this this system, at first, the output from an array of piezoelectric sensors is fed to the ac ripple neutralizer. A constant output voltage irrespective of fluctuations will be maintained by a voltage regulator. This regulated voltage is stored in the battery and is fed to the microcontroller. The LCD which is interfaced with the microcontroller in land displays the amount of charge stored by the battery. In this system the power generated has been used for applications such as charging E- vehicles. A buzzer is used to alert when the battery voltage falls below the required voltage for charging the microcontroller.



CHAPTER 6

FUTURE AND SCOPE

Growing Population which is considered to be a bane is used advantageously with piezo application. In an era where the significance of renewable energy is well proved, piezoelectricity is a promising alternative source of energy. At present the usage of piezoelectricity seems far from practicality for portable gadgets. But designers concerned about the energy crisis are working to enhance its utility for portable gadgets and beyond. Piezoelectricity shall definitely have huge impact in near future. At present it finds numerous applications, which will increase further owing to the desirable characteristics of the technology.

Maximum crowded public can be seen in railways stations, airports, metro station and holy places. Such places can be use of piezoelectric crystals for generation of electricity. In holy places gathering ranging from thousands to millions, so setting up of piezoelectric crystals at floor would generate enough power to light up lights of houses. Use of piezoelectric crystals has been started and hopeful results are attained. Better manufactured piezoelectric crystals and better selection of place of installations with future advancement in field of electronics so more electricity can be generated and it can be viewed as a next auspicious source of generating electricity,

Human have needed and used energy at an increasing rate for their sustenance and well-being ever since they came on the earth a few million years ago. Due to this a lot of energy resources have been exhausted and wasted. Proposal for the utilization of waste energy of foot power with human locomotion is very important for populated countries like India and China.

We can use this principle in the speed breakers at high ways, where the vehicles rush too much, thus increase input torque and ultimate output of generator

If we improve piezoelectric material, then output efficiency can be increased by using large sized piezoelectric transducers which in turn gives a large amount of power.

If this mechanism is implemented on large industrial scale, then output Wattage can

be enlarged Like Footpaths and Roads Etc.

In some of methods are also utilized for energy generation by various methods i.e.

- Trade-mill.
- Smart Street light.
- Piezoelectric road.
- Piezoelectric Stairs.
- Dancing floor.
- Flooring tiles etc.

6.1 TREADMILL

All of us is well known about that 'Health is wealth' and here we are making it as "There is no way to step off the tread mill. You have to keep walking constantly to stay fit and generate electricity for this world".

We are using the trade mill which have the piezoelectric ceramic pieces are embedded in a running platform below a running belt to be fixed. The storage battery is mounted in the middle of the running platform. The insulating layer is wrapped outside the positive wires and the negative wires led out of the piezoelectric ceramic pieces. The positive wires and the negative wires which the insulating layer is wrapped outside are connected to the storage battery. Electric energy stored in the storage battery can be supplied to the treadmill for movement.



Figure No (xvii)

6.2 SMART STREET LIGHT

Has been a while since artificial light has allowed us to see in the dark. We can now light the home we live in, our workspace and of course, the streets we wander. This invention has made lives easier and, as public lighting has become a norm in most parts of the world, safer. Street lighting is a core piece of urban and rural infrastructure, creating a safe environment for pedestrians and drivers alike.

The street light system using a Piezoelectric transducer in the area of science and technology power source for movable devices and wireless sensor network systems. Electricity has become one of the main roles in current days, so. The Smart Street Light supplies a solution for energy saving

By utilizing Piezoelectric, it affects Piezoelectric gadget transmute that energy into present and storing that energy in rechargeable battery, so that controlling street light during day hours. without human involvement.

Smart street lighting is the backbone for smart cities of the future. Connecting over 360 million streetlights worldwide, smart street lighting systems turn cities lighting grid into one centrally controlled network. With non-stop access to power, street poles are ideal for mounting smart city systems such as security cameras, environmental sensors, traffic counters or electric vehicle chargers. By connecting every lighting pole to a wider network connection, each lamp becomes a platform which will act as a binder for wider smart city investments.



Figure No (xviii)

6.3 PIEZOELECTRIC ROAD

Piezoelectric sensors collect data by converting mechanical energy into electrical energy. When used to count vehicles, the sensor is mounted in a groove cut into road's surface. When a car drives over the piezoelectric sensor, it squeezes it and causes an electric potential signal. The size of the signal is proportional to the degree of deformation. When the car moves of the voltage reverses. This change in voltage can be used to detect and count vehicles. The counting device which is connected to the sensors is housed in an enclosure by the side of the road.



Figure No (xix)

6.4 PIEZOELECTRIC STAIRS

A person exerts lots of force when they walk down the stairs. The staircase power harvesting system in tends to turn this energy into electrical power using a pico-electric generator.

This study is a part of a research series conducted to design a sustainable energy-harvesting fairway. This research proposes a semi-public energy-harvesting stairway which is designed to he used by the public and to be placed both outdoor and indoor. It could be installed inside lecture halls, hotels, footbridges and etc. This characteristic has been included incorporated into the stairway's whole design and product specifications. After this study had been conducted, a mock-up was created and tested to gain knowledge in terms of electrical input, output, and the chronological process in producing electricity.

Spurway's product description had been acquired; it had to be transformed into a finalised design. In general, the design is made up of three main elements i.e.

- The Frame.

- The Stairs, and
- The Generator Set.

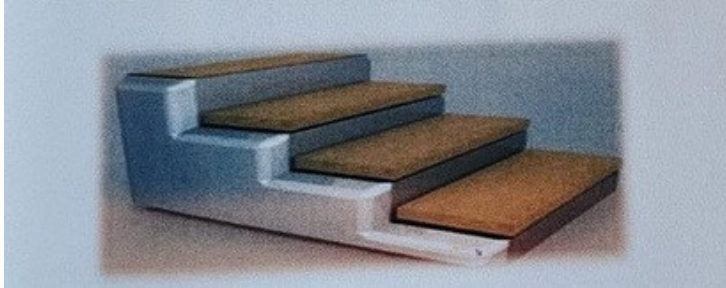


Figure No (xx)

To ensure the its structure strong and robust, the frame is designed to be made of steel. Furthermore, the trestle structured frame would distribute weight equally, thus keeping it standing firm even if loaded with a heavy weight. The foothold moves relatively to the frame. A knob is placed beneath each foothold to channel users' pressure force while moving downwards as the foothold is being stepped thus spinning the gear placed beneath. Between the foothold.

the section of the frame which supports the foothold, an empty space is provided to allow the foothold moves vertically when pressurized the vertical movement of the foothold is enabled by the spring coil placed at the empty space between the bottom side of each foothold and the upper side of the frame which supports it. The distance of the movement is determined by the pressure which the spring coil bear. In addition to several spring coils, piezoelectric sensors and sponge reducers are also installed. The piezoelectric sensor acts as a secondary power producer while the generator system acts as the primary one. The sponge reducers are installed to stabilize the foothold while being stepped on.



Figure No (xxi)

6.5 DANCING FLOOR

Europe is another one of the Country which started experimenting use of piezoelectric crystal for energy generation in night clubs. Floor is compressed by the dancer's feet and piezoelectric materials makes contact and generate electricity. Generated electricity is nothing but 2-20 watt. It depends on impact of the dancer's feet. If constant compression of piezoelectric crystal causes a huge amount of energy.

See how this works when mechanical movement (like walking Let's see or dancing) is involved. If you happen to dance on a floor that is equipped with piezoelectric technology, as soon as your feet hit the floor, a certain amount of pressure is applied. This pressure is captured by sensors present in the floor, and is subsequently converted into an electrical charge by piezo materials usually presenting the form of crystals or ceramics).

This charge is then stored and can later be used as a power source.

In order to harness viable energy using this method, it is crucial that there be a large number of people on the floor who are in a state of movement at the same time. In other words, you should be looking for a crowd, like the ones you see at busy railway stations and shopping malls. You need a crowd because a single footstep doesn't provide enough pressure, but when a large number of people are walking, running or dancing at the same time, it makes a huge difference. To be more specific, you need approximately 28,500 simultaneous footsteps to power a train for one second.



Figure No (xxii)

6.6 FLOORING TILES

Japan has already started experimenting the use of piezoelectric effect for energy generation. They implement piezoelectric effect on the stairs of the bus. Thus, every time passenger steps on the tiles; they trigger a small vibration that can be stored as energy.

The flooring tiles are made up of rubber which can absorb the vibration. This vibration generates when running or walking on it. Under these tiles piezoelectric material is placed. When the movement is felt by the material, they can generate the electricity. This generated energy simultaneously stored into the battery. Generated electricity we can use the lightning of lamp or street light. Energy is generated by step of one human being is too less but if number of steps increases ultimately energy production also increases.

A piezo tile capable of generating 12V has been devised. it was established that series-parallel combination connection is more suitable. The weight which is applied to the piezoelectric tile and corresponding voltage generated is studied and they are found to have a linear relation. It is especially suited for implementation in crowded areas. This can be used in street lighting without the use of long power lines. It can also be used as charging ports, a lighting of pavement side buildings.



Figure No (xxiii)

CHAPTER 7

CONCLUSIONS

Throughout the course of this project, we have explored, designed, and successfully implemented a system that harnesses the piezoelectric effect—a phenomenon where certain materials generate an electric charge in response to mechanical stress. By strategically placing piezoelectric sensors in walkways, footpaths, or areas of high pedestrian traffic, this system captures energy produced from the pressure of human footsteps and converts it into electrical energy. This generated energy is then stored in a rechargeable battery or capacitor and can be used to power low-voltage devices, such as LED lighting, digital counters, or small sensors.

The integration of smart electronics with this energy harvesting system has further enhanced the intelligence and utility of the design. Microcontrollers were used to monitor energy production, manage energy storage, and control output applications. The data collected through these smart systems can also be utilized for crowd analysis, smart city planning, and environmental monitoring, adding an entirely new layer of functionality and societal benefit.

This system is not only a technological innovation but also a socially relevant and eco-friendly solution. It demonstrates how renewable micro-energy generation can be seamlessly embedded into our daily lives without requiring any behavioral change. Unlike traditional energy generation systems, which depend on fossil fuels or large infrastructure, the smart footsteps power generation system is decentralized, low-cost, maintenance-friendly, and sustainable.

From an academic and engineering perspective, this project allowed us to gain practical experience in multiple domains, including sensor technology, energy conversion, embedded system design, circuit implementation, and real-time monitoring. It also encouraged innovative thinking and problem-solving in addressing real-world challenges using interdisciplinary knowledge.

Furthermore, the scalability and adaptability of this system make it highly promising for future development. By using more efficient piezoelectric materials, advanced power management ICs, and improved mechanical designs, the energy output can be significantly increased. The system can also be networked across large areas—such as railway stations, airports, shopping malls, and stadiums—where thousands of footsteps daily can contribute to a meaningful amount of energy generation.

In conclusion, the Smart Footsteps Power Generation Using Piezoelectric Sensor is not just a conceptual project but a prototype of a practical and impactful technology that aligns with the global vision of sustainable energy development and smart infrastructure. With further optimization and mass implementation, this system holds the potential to

revolutionize how we think about energy, turning every footstep into a source of power, and every person into an active contributor to clean energy generation.

As a fact only 11% of renewable energy contributes to our primary energy. If this project is deployed then not only, we can overcome the energy crises problem but this also contributes to create a healthy global environmental change.

- Smart system.
- Produce 200 watts of electricity.
- Durable
- Have a life of approx. 5 years.

This technique produces electricity with the assistance of electricity components that create use of the energy of human footsteps. The potency of the total system is measure employed in place of the standard ones then a lot of charge will be hold one than the standard ones. It stores and discharge energy while not intense abundant energy. Thus, the need of constant increase of power will be met by putting in these systems in heavily packed places. This may doubtless not solely overcome. The energy crises however conjointly build up a healthy encompassing.

Use of piezoelectric crystals has been started and positive results are obtained. With further advancement infield of electronics, better synthesized piezoelectric crystals and better selection of place of installations, more electricity can be generated and it can be viewed as a next promising source of generating electricity.

Regardless of this project, the future of piezoelectric materials looks bright, with studies focusing on their properties and applications. If a compromise between the hardness of the road and the make-up of the small devices is reached, then undoubtedly the system will benefit both drivers and the national power grid. The assembly developed using series and parallel combination of piezo crystals is very cost effective.

A single crystal costs around 23-25 Rupees, and hence the cost of whole assembly is very less. It is very encouraging to get a good voltage and current at such a low cost at the same time utilizing the waste energy. So, the assembly improves on the concern of cost effectiveness to a great extent and the work is on to further improve upon the results of the system

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